Agricultural Water Management 210 (2018) 224–231

Contents lists available at ScienceDirect

Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat

Effect of irrigation regime on yield, harvest index and water productivity of soybean grown under different precipitation conditions in a temperate environment

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ARTICLE INFO

Keywords:
Evapotranspiration
Agricultural water management
Glycine max (L.) Merr.
Water productivity
Yield composition

ABSTRACT

In temperate climatic regions, agricultural production depends on precipitation amount and its distribution during the growing season. A 3-year field study was conducted to investigate the effects of different irrigation regimes on yield parameters and water productivity of sprinkler-irrigated soybean [*Glycine max* (L.) Merr.], grown under wet, semi-dry and dry conditions in a temperate environment. Four irrigation levels were applied: full irrigation (I_{100}), 65% and 40% of full irrigation (I_{65} and I_{40}) and non-irrigated control (I_0). On average, the I_0 treatment resulted in the highest harvest index (HI) and I_{100} produced the lowest HI. A significant quadratic correlation between seed yield and crop water use was observed in dry and semi-dry year. The irrigation regime significantly influenced seed yield and water use. I_{65} treatment produced the highest seed yield (3.69 t ha^{-1}) and showed the highest water productivity (WP) (0.90 kg m^{-3}) and irrigation WP (1.08 kg m^{-3}). The present study indicated that irrigation is necessary for soybean cultivation in semi-dry and dry years i.e., when seasonal precipitation is lower than about 300 mm. In wet years, with a favourable amount and distribution of precipitation during the growing season, yields are similar to those achieved with irrigation and high ET values of soybean are attributable to increased evaporation.

1. Introduction

Temperate climate is characterized by large variability of weather parameters. At most places, agriculture depends on precipitation. Its amount and distribution vary from year to year and during growing seasons, which causes instability of production and reduces farmers' revenue. Therefore, supplemental irrigation is needed to increase and stabilise agricultural production especially during the precipitation scarce seasons.

Vojvodina is located in a temperate climatic zone of the Pannonian Plain, in the northern part of the Republic of Serbia. Agricultural production is prevalently rain-fed although it is one of priority economic sectors. Nevertheless, in this region, as in many similar regions around the world, the availability of freshwater resources for agricultural use has been decreasing due to climate change and increasing demand of other sectors.

Drought affects soybean growth and productivity in many parts of the world (Hatfield and Prueger, 2011; Steduto et al., 2012; Sentelhas et al., 2015). Water stress is principally harmful during flowering, seed setting and seed filling. The research findings indicated that limited, supplemental irrigation during the course of growing season or in a specific growth stage can significantly increase water productivity (WP) and soybean yields (Giménez et al., 2017; Jha et al., 2018). Response to irrigation depends on climate, rainfall pattern during growing season, soil properties, cultivar, agronomic practices, and experimental procedures.

Limited irrigation can result in substantial differentiation of crop productivity in various environments (Djaman et al., 2013). Akcay and Dagdelen (2016) reported that deficit irrigation may support the water saving irrigation programs under conditions of water scarcity. They stated that effective on-farm implementation of deficit irrigation might improve crop productivity when the data about crop yield response to water inputs are available. Knowledge of crop response to water stress over the growing season is useful also in crop growth modelling, enabling both improved prediction of crop performance in different situations and adoption of water saving management practices.

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https://doi.org/10.1016/j.agwat.2018.08.002
Received 4 April 2018; Received in revised form 1 August 2018; Accepted 3 August 2018
Available online 24 August 2018

In the Vojvodina region, soybean is one of the main field crops grown prevalently under rain-fed conditions. In the recent years, irrigation is becoming a common practice due to variable precipitation pattern and frequent drought caused by climate change. However, information is lacking on the impact of different irrigation levels on soybean yield and water productivity under different precipitation patterns. Consequently, the objective of this study is to compare the effects of three irrigation treatments and non-irrigated treatment on the growth, yield and water productivity of soybean grown under different precipitation conditions in a temperate climatic region. These data can be useful for soybean industry and the regional soybean growers to maximize the seed yield and productivity of water use through the selection of appropriate irrigation scheduling strategy.

2. Materials and methods

2.1. Site description and experiment set up

Field experiments were carried out during three growing seasons (2006, 2007, 2008) at Maize Research Institute "Zemun Polje", Vojvodina, Serbia (44°52′ N, 20°20′ E, 81 m above sea level). The soil is classified as silty loam Calcaric Chernozem up to 0.60 m. Plant available water holding capacity to this depth (difference between –0.03 and –1.5 MPa) is 118 mm. The main chemical and physical characteristics of the soil in the study area are shown in Table 1.

The experimental area is characterized by a mild temperate climate with hot and dry summers. The long-term 30-years average annual precipitation is 638 mm with a notable inter- and intra-seasonal variability in amount and distribution pattern. The long-term average air temperature is 11.9 °C. The air temperature, precipitation, humidity and wind speed data were taken from Surčin Meteorological Station (Table 2). Rain-fed farming and occasional irrigations are the main production system in the study area.

The experiment was arranged in a randomized complete block design with four treatments and four replications. The following irrigation regimes were applied: I_{100}, I_{65}, I_{40}, and I_0, closely corresponding to no water deficiency treatment (full irrigation), 65% of full irrigation (35% deficit), 40% of full irrigation (60% deficit) and non-irrigated (rain-fed). In I_{100} treatment, the irrigation events replenished fully water depleted by evapotranspiration up to the field capacity (FC). In I_{40} and I_{65} treatments, the irrigation amounts referred to approximately 40% and 65% of I_{100} amount. The irrigation rates and timings were determined following the soil water content in the effective root zone depth (going up to 0.60 m). In all irrigated treatments, irrigation was practiced when soil water content reached approximately 50% of total available water (the difference in soil water storage between field capacity and wilting point). Full irrigation treatment includes the reservoir fulfillment up to water content at field capacity, whereas in the treatments I_{65} and I_{40}, 65% and 40% of full irrigation amounts were applied. The irrigation rates and time were determined depending on the soil water content in the effective root zone depth (to 0.60 m). Irrigation events were stopped two to three weeks before harvesting.

Irrigations were applied with the hand-move sprinkler irrigation system. Sprinklers heads with two nozzles mounted at 2.0 m were organized in 12 m × 12 m square grid. Operating pressure was about 3.50 MPa with a discharge of 0.41 l s^{-1}. Irrigation started on 11 July and 10 July in the first two seasons (2006 and 2007, respectively), and on 4 June in the third season. The last seasonal irrigations were applied in the third decade of July, and in the third and second decade of August in the first, second and third growing season, respectively.

Each experimental plot was designed 7.14 m long and 8.0 m wide (57.12 m^2). To minimize irrigation edge effects, the inter-plot space was 2.0 m.

Soybean cv. Nena (maturity group II) obtained from the Maize Research Institute "Zemun Polje" Company, was sown on 28 April 2006, 24 April 2007 and 6 May 2008 in 0.50 m-wide rows at a seeding rate of 4.45 seed m^{-2}. The crop was fertilized at the recommended rate, based on soil tests, with 450 kg ha^{-1} of NPK (15:15:15) before sowing, and a further nitrogen dose of 69 kg ha^{-1} was added as urea, when the plants were 25–30 cm in height. Weeds, pests and diseases were adequately controlled. Only the irrigation management differed between the treatments.

2.2. Crop and yield sampling

Seed yield was measured by harvesting an area of 5.0 m^2 per plot. Hand-harvest took place on 19, 18, and 15 October, respectively for 2006, 2007 and 2008. Seed yield was normalized for 13% seed water concentration. Harvest index was determined as seed yield divided by the total aboveground biomass after drying the samples at 65 °C.

2.3. Soil water and crop evapotranspiration

Soil water content was measured periodically throughout the growing seasons using gravimetrical method at different depth, starting at 0.10-m at sowing and going down to 0.60 m at flowering and hence after. The measurements were based on the conventional oven-dry mass. Gravimetrical water content was converted to volumetric water content after multiplying it by the bulk density. Soil bulk density was measured at the beginning of each crop season at 0.10 m interval depths up to 0.60 m.

Actual crop evapotranspiration or total water consumption (ETa, mm) of each plot was calculated according to Sincik et al. (2008):

$$ETa = P + I + C_r - D_p - R_f \pm \Delta S \quad (1)$$

where P is the rainfall in the growth period (mm), I is the amount of net irrigation water applied to individual plots (mm), Cr is capillary rise of water from below the root zone (mm), Rf is the surface runoff (mm), Dp is the water lost through deep percolation (mm), and ΔS is the cumulative change in the soil water storage between planting and harvest within the effective crop root zone (mm). Irrigation scheduling was based on the soil water balance method and gravimetric soil water content measurements. In Eq. (1) *Cr* was assumed to be zero, because the water table was at 80 m depth. Deep percolation was ignored because soil water content measurements showed that no water percolated below 100 cm depth. The run-off and run-on was also assumed to

Table 1
Physical and chemical properties of the Calcaric Chernozem at the experimental site.

Soil depth (cm)	Particle size distribution (%)			Bulk density (g cm^{-3})	Field capacity (% vol.)	Wilting point (% vol.)	Available water (% vol.)	pH (1:2.5 H_2O)	$CaCO_3$ (%)	Organic matter (%)	Available nitrogen (mg kg^{-1})	P_2O_5 (mg kg^{-1})	K_2O (mg kg^{-1})
	Clay[a]	Silt	Sand										
0–20	28.4	67.3	4.3	1.21	31.1	15.0	16.2	7.73	4.3	2.99	27.5	660	250
20–40	26.7	69.2	4.2	1.33	37.0	15.1	21.9	7.77	5.6	2.92	39.9	380	200
40–60	26.6	68.3	5.2	1.28	37.5	14.9	22.6	7.92	22.7	1.57	21.1	85	80

[a] Clay < 0.002 mm, Silt 0.05–0.002 mm, Sand 2.00–0.05 mm.

Table 2
Average of daily values of air temperature (T_a), relative humidity (RH), wind speed at 2-m height (u_2), and precipitation (P) for the month of April–October at Zemun Polje, Serbia in 2006–2008 crop seasons, and perennial averages (1981–2002).

Months	T_a (°C)				RH (%)				u_2 (m s^{-1})				P (mm)			
	2006	2007	2008	20-yr avg.	2006	2007	2008	20-yr avg.	2006	2007	2008	20-yr avg.	2006	2007	2008	20-yr avg.
April	12.2	13.2	13.4	12.0	72.2	50.4	63.8	68.0	1.51	1.71	3.19	3.34	93	31	27	60
May	15.8	19.0	18.3	17.2	67.6	63.7	56.8	66.5	1.34	1.94	2.41	2.84	33	42	40	56
June	18.8	22.5	22.3	20.1	76.1	55.8	59.1	70.0	1.17	1.38	2.34	2.49	144	63	36	95
July	22.8	23.9	22.6	21.8	62.0	47.0	55.1	67.0	1.07	1.57	2.76	2.30	27	19	46	57
August	19.6	23.7	22.8	22.2	77.7	59.7	54.2	66.0	1.09	1.62	2.35	2.21	109	52	20	62
September	18.6	15.0	16.6	18.0	73.8	68.4	67.8	72.0	1.25	1.81	2.26	2.34	11	73	55	55
Seasonal average/total	18.0	19.5	19.3	18.6	71.5	57.5	59.5	68.3	1.24	1.67	2.55	2.59	417	279	225	384
Anual	11.4	12.9	12.3	11.9	74.8	66.8	62.8	74.13	1.79	1.83	2.67	2.82	661	670	486	638

be zero, because the field was flat and the irrigation volume per application was less than or equal to FC.

2.4. Yield and water productivity assessment

Water productivity (WP) (kg m^{-3}) and irrigation water productivity (IWP, kg m^{-3}) were determined to evaluate the productivity of different water regimes. These indices were calculated as (Sincik et al., 2008):

$$WP = Y/ETa \tag{2}$$

$$IWP = (Yi - Yo)/I \tag{3}$$

where Y is the seed yield (kg ha^{-1}), ETa is the actual crop evapotranspiration (mm) calculated from Eq. (1), Yi is the irrigated treatment seed yield (kg ha^{-1}), Yo is the rain-fed seed yield (kg ha^{-1}) and I is the seasonal amount of irrigation water applied (mm).

2.5. Statistical analysis

Data were subjected to an over-year analysis of variance (ANOVA) as a Randomised Complete Block design with four replications. Analysis was conducted using IBM SPSS 20 statistical package and the means were compared using the Fisher's least significant difference (LSD) test at 5% significance level. The relationship between seed yield and crop evapotranspiration (ETa) was evaluated using regression analysis.

3. Results and discussion

3.1. Weather conditions during the growing period

The first growing season was cooler and wetter than the other two seasons (Table 2). The average temperature during the first growing season was slightly lower than the long-term mean temperature of 18.6 °C. The average temperatures during the second and third seasons were higher than the long-term mean temperature. The three investigated growing seasons were characterized as wet (season 2006), dry (season 2007) and very dry (season 2008) with 108, 73, and 58% of the long-term mean precipitation, respectively. The first growing season was wet, with rainfall > 100 mm in June and August, much higher than the long-term average. Rainfall was unevenly distributed and irrigation was applied in each year, except in the first season, in the I_{40} treatment (Table 3).

3.2. Applied irrigation water and seasonal water consumption

The total amount of irrigation water varied depending on the seasons and irrigation treatments. The differences between three seasons were due to change in the amount and pattern of precipitation (Table 3). In the first growing season, rainfall was particularly abundant in June. Full irrigation was only 60 mm, i.e., it was more than two times less than in 2007 (135 mm) and almost four times less than in 2008 (235 mm). A similar difference in the total amount of irrigation water applied in different years was observed also for other treatments. The differences were due to the amount and pattern of precipitation (Fig. 1). In general, the lower humidity and higher air temperature resulted in greater demand of water for soybean crop.

The seasonal crop evapotranspiration (ETa) increased with the increase in the irrigation volume (Table 3). The highest soybean seasonal ETa (461 mm on average) was observed in the I_{100} treatment. Other treatments, subjected to severe water deficits, had lower ETa. The lowest ETa was recorded in the rain-fed treatment (326 mm on average). High precipitation in the first growing season limited irrigation inputs and did not allow a wide variation of ETa, which ranged from 440 mm in rain-fed to 505 mm in full irrigation treatment. Similarities in ETa between irrigation treatments were found for the second and the third growing season. The ETa was the lowest in the dry (2008) growing season ranging from 227 mm in rain-fed to 450 mm in full irrigation treatment.

According to Doorenbos and Kassam (1979) estimated ETa for maximum production in soybean is between 450 mm and 700 mm depending on the length of growing period, cultivar, soil conditions and climate. Sincik et al. (2008) observed soya bean ETa of 821, 471–634, and 373 mm for full irrigated, deficit irrigated, and dry-land plants, respectively, in the conditions of Bursa, Turkey. Suyker and Verma (2009) reported ETa of 431 mm and 452 mm for rain-fed and irrigated soybean, respectively, in eastern Nebarska.

Table 3
Evapotranspiration (ETa) and seasonal applied water.

	Seasonal total ETa (mm)				Seasonal applied water (mm)			
	2006	2007	2008	3-year	2006	2007	2008	3-year
I_0	440	308	227	325	0	0	0	0
I_{40}	442	326	330	366	0	54	94	49
I_{65}	478	368	384	410	39	88	153	93
I_{100}	505	427	450	461	60	135	235	143
X (overall mean)	466	357	348		50	92	161	

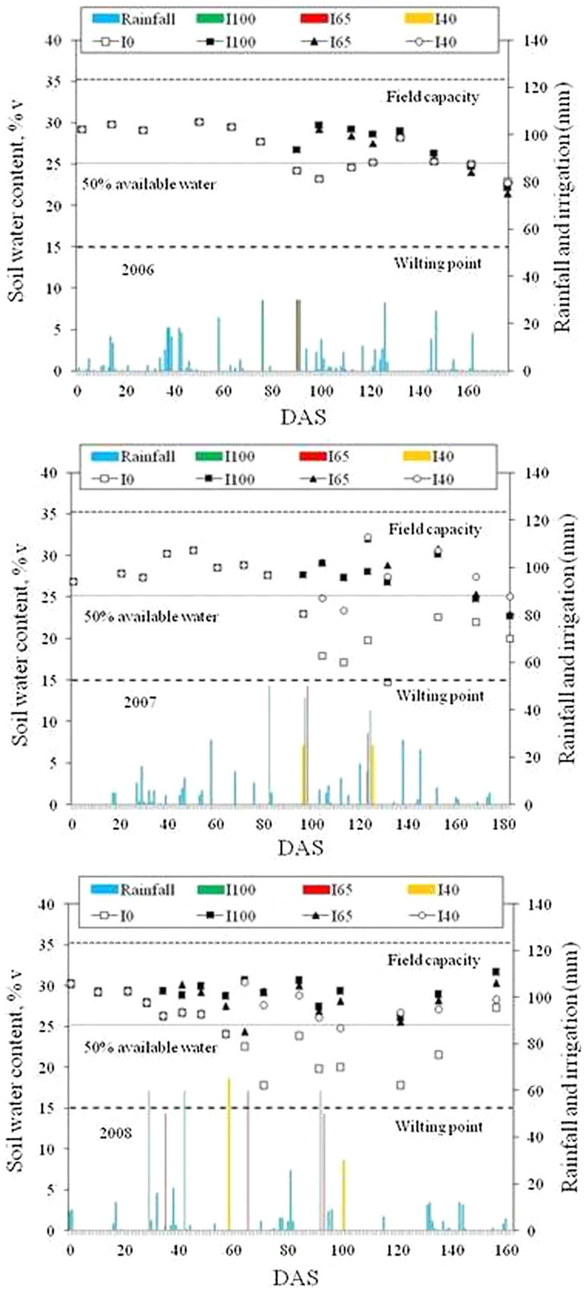

Fig. 1. Volumetric soil water content in the crop root zone (0–0.60 m) for different irrigation regimes during three soybean growing seasons (DAS – days after sowing).

3.3. Soil water content

Fig. 1 shows the soil water content (%) dynamics of each treatment in the top 0.60 m of the crop root zone during three growing seasons of soybean. The differences between the treatments augmented with the reduction of seasonal precipitation and with the increase of irrigation requirements. In the case of irrigated treatments, the soil water content was almost always above the threshold level fixed to 50% depletion of total available water. The greatest variation of soil water content was observed for I_0 treatment. In wet year, soil water content of rain-fed treatment was similar to the irrigated treatments except a slight deviation in July, which corresponded to flowering and pod formation periods. In other two years, water stress started earlier, it was prolonged for more than 60 days, and soil water content has reached almost the level of wilting point.

3.4. Aboveground biomass yield

As expected, the I_0 treatment had the lowest aboveground biomass yield due to water stress, which was especially severe in dry, third, season (Table 4). No significant differences in aboveground biomass yield were found between I_{40} and I_0 treatments in wet, first, season. The lower irrigation amount for the I_{40} treatment contributed to lower aboveground biomass compared to I_{65} and I_{100}. The results indicated that lower irrigation level in I_{65} and I_{40} treatments significantly inhibited soybean growth if compared to that in I_{100}. In irrigated and rain-fed treatments, the aboveground biomass varied from 8.09 to 11.64 t ha^{-1} and from 5.91 to 8.00 t ha^{-1}, respectively. Overall, the aboveground biomass yields were 2–73% greater in irrigated compared to non-irrigated plots, depending on the growing season and treatment. A similar effect of irrigation on aboveground biomass yield was observed by other researchers (Karam et al., 2005; Sincik et al., 2008; Jha et al., 2018).

3.5. Seed yield

Results obtained in this study showed that irrigation treatments significantly affected seed yield in each of three growing seasons (Table 4). The seed yields were the highest during the first - wet season in respect to two drier growing seasons.All irrigated treatments significantly increased yield compared with the rain-fed treatment. Therefore, limited precipitation and the reduced soil water content were the main factors limiting seed yields of soybean in the Vojvodina region. On average, treatment I_{65} had the greatest seed yield (3.69 t ha^{-1}) followed by I_{100}, and I_{40} irrigated treatments with 3.32, and 2.97 t ha^{-1} respectively. The possible factors involved in low yield under high irrigation treatment are excessive crop growth during the vegetative stage and poor soil aeration (Huang et al., 2002). The I_0 treatment had the lowest seed yield. Soybean yields were the most responsive to irrigation in the third, dry, growing season. The triennial impact of irrigated treatments on yield, led to an increase in seed yield by 28%, 42%,

Table 4
Yield components of soybean under different irrigation regimes in the three growing seasons.

Irrigation treatments	Above-ground biomass (t ha^{-1})				Seed yield (t ha^{-1})				Harvest index (%)			
	2006	2007	2008	3-year	2006	2007	2008	3-year	2006	2007	2008	3-year
I_0	8.00b†	7.66d	5.91d	7.19	3.26c	2.88c	1.63d	2.59	0.407a	0.376a	0.276d	0.353
I_{40}	8.09b	9.77c	8.63c	8.83	3.27c	3.09b	2.55c	2.97	0.404a	0.316b	0.295b	0.386
I_{65}	10.71a	11.64a	10.21a	10.85	4.27a	3.60a	3.21a	3.69	0.399a	0.309b	0.314a	0.341
I_{100}	10.43b	10.50b	9.39b	10.11	3.77b	3.50a	2.69b	3.32	0.362b	0.333b	0.287c	0.327
X (overall mean)	9.31	9.89	8.53		3.64	3.27	2.52		0.393	0.334	0.293	0.352
Year (Y)				**				**				*
Y x I				**				**				**

†In each growing season, values followed by different letters are significantly different among treatments according to LSD test at P ≤ 0.05.

and 15% for I_{100}, I_{65}, and I_{40}, respectively (average 29%), compared with the I_0 (non-irrigated) treatment. Although I_{65} irrigation treatments received about 50, 41, and 32% less irrigation water in respect to the I_{100} plots, the seed yield was increased by an average of 13.3, 2.9, and 19.3% in 2006, 2007 and 2008 season, respectively, as compared to I_{100} (Table 4). There were no statistically significant differences in seed yield between the I_{40} and I_0 in the first, wet, year, because of adequate precipitation during the growing season. In addition, there were no statistically significant differences between the I_{100} and I_{65} treatments in the second growing season. Consequently, the difference in seed yield between the growing seasons could result from the rainfall amount and distribution, and elevated air temperatures and increased evapotranspiration demand.

The results of present study are in agreement with those reported by Eck et al. (1987) and Karam et al. (2005). They have found significant decrease in seed yield due to water stress. Sincik et al. (2008) reported a significant effect of limited irrigation on seed yield of soybean. Koivisto et al. (2003) stated that significant differences in soybean performance between two experimental years were due to differences in the environmental conditions of the United Kingdom. The findings of present study do not correspond to those of Payero et al. (2005), who reported that irrigation treatments had only statistically significant effect on the yield in the two driest seasons (2002 and 2003) under semi-arid conditions of west-central Nebraska.

3.6. Harvest index

Harvest index (HI) had irregular variation from I_{100} to I_{40} plots (Table 4). Irrigation regimes had significantly different effects on HI in all three growing seasons but there was no significant difference between the irrigated treatments in the second season and between the I_{65} and I_{40} treatments in the first, wet, growing season. On average, the I_0 treatment resulted in the highest harvest index and I_{100} produced the lowest HI. In the present study, harvest index increased slightly, on average, while the seasonal irrigation volume decreased. HI of irrigated treatments was 3–7% lower compared to I_0. Sincik et al. (2008) observed irregular variation of HI and reported that HI tended to be higher in non-irrigated treatments. Pedersen and Lauer (2004) found that irrigation lowered HI by 2%, on average. In contrast to the present study, Garcia y Garcia et al. (2010) found that different irrigation regimes did not affect harvest index of soybean in a humid region of the south-eastern USA. In addition, Demirtas et al. (2010) stated that HI of drip-irrigated soybean was not affected by drought stress in a sub-humid environment of Turkey.

3.7. Water productivity and irrigation water productivity

The WP values for sprinkler-irrigated soybean varied considerably with irrigation treatment in all three experimental seasons, especially during the driest, third, growing season (Table 5). The WP of irrigated treatments did not show consistent difference with decreasing applied irrigation water. The WP of I_{65} was the highest in all seasons and it was, on average, greater by 13% than WP of non-irrigated treatments. However, the I_{100} treatment had a WP value ~10% lower than the non-irrigated one. The WP tended to be higher in moderately humid second season. No significant difference was found between non-irrigated treatment and I_{100} and I_{40} treatments in the first, wet, growing season. The I_{65} treatment resulted in the highest WP in all three growing seasons, which means that, on average, in the I_{65} treatment more seed mass was produced per mm of water, compared with the I_{40} and I_{100} treatments. The WP of I_{100} and I_{40} were, on average, 24% and 14% less than in the I_{65} treatment, respectively. These results are in agreement with those of Burriro et al. (2002) and Sincik et al. (2008).

Table 5
Water productivity (WP) and irrigation water productivity (IWP) of soybean.

Irrigation treatments	WP (kg m^{-3})				IWP (kg m^{-3})			
	2006	2007	2008	3-year	2006	2007	2008	3-year
I_0	0.74b[†]	0.94a	0.73c	0.80	0.00	0.00	0.00	0.00
I_{40}	0.74b	0.93a	0.77b	0.81	0.00	0.42b	0.97a	0.69
I_{65}	0.89a	0.98a	0.84a	0.90	3.35a	0.90a	0.98a	1.74
I_{100}	0.75b	0.82b	0.60d	0.72	0.85b	0.46b	0.45b	0.59
X (overall mean)	0.78	0.916	0.73		2.1	0.59	0.80	
Year (Y)				**				**
Y x I				**				**

[†]In each growing season, values followed by different letters are significantly different among treatments according to LSD test at P ≤ 0.05.

On the contrary, Sincik et al. (2008) and Demirtas et al. (2010) reported remarkably lower WP values, from 0.46 to 0.58 kg m^{-3} and from 0.41 to 0.64 kg m^{-3}, respectively, of deficit irrigated soybean in a sub-humid region of Turkey. Katerji et al. (2008) quoted WP values of 0.39–0.77 kg m^{-3} for three studies in Mediterranean environment. Only the results obtained by Katerji and Mastrorilli (2009), with WP values of 0.73 (clay soil) and 0.81 kg m^{-3} (loam soil), are comparable to those of the present study. Nevertheless, Eck et al. (1987) observed that applying water deficit did not increase seasonal water productivity of soybean, in environmental conditions of the Southern High Plains of the USA. They also noted that moderately stressed or unstressed treatments had higher WP than those with severe water stress.

Irrigation treatments influenced IWP much more than WP. In the present study, IWP showed irregular variation from I_{40} to I_{100} but tended to be higher in I_{65} treatment (Table 5). There was significant variation in IWP between experimental seasons. The difference of average IWP between the treatments in the same growing season was smaller than that between the seasons in the same treatment. Large variation of IWP was found as a result of significant differences in applied irrigation water across growing seasons. Therefore, the relatively wet first season resulted in significantly higher IWP values, compared to the other two growing seasons. The lower IWP in the last two seasons may be attributed the lower precipitation than in the wet season.

The lowest IWP was obtained from treatment I_{100} in all three growing seasons. The IWP values observed in the I_{65} treatments were significantly higher than those observed in I_{100} and I_{40} treatments in all three growing seasons. Hence, it can be concluded that I_{65} irrigation could be adequate strategy for soybean production in the Vojvodina region.

The IWP values from this study were higher than those reported in many other studies (Evett et al., 2000; Sincik et al., 2008; Demirtas et al., 2010). However, the IWP values observed in this study are in agreement with those of Garcia y Garcia et al. (2010), who reported IWP for sprinkler irrigated soybean, ranging from 0.55 to 1.15 kg m^{-3}, in a humid region of USA. Indeed, it has often been considered that irrigation in humid zones is more efficient than irrigation in the arid regions, because the lower water vapour deficit of the humid regions results in higher transpiration efficiency than in the arid climates (Tanner and Sinclair, 1983).

3.8. Relationships between seed yield and seasonal water consumption

The results of regression analysis between the seed yield and seasonal crop evapotranspiration (ETa) showed quadratic relationship (R^2 = 0.6486) for all irrigation regimes and for all three growing seasons (Fig. 2). Therefore, there is an optimal water application amount to maximize the effectiveness of irrigation on increasing grain yield above the rain-fed yields.

Similarly to these results, an exponential relationship between soybean seed yield and seasonal ETa was observed by Demirtas et al. (2010) in the sub-humid environmental conditions in Turkey(). In contrast to the present study, linear relationships between crop evapotranspiration and soybean yield were reported by Payero et al.

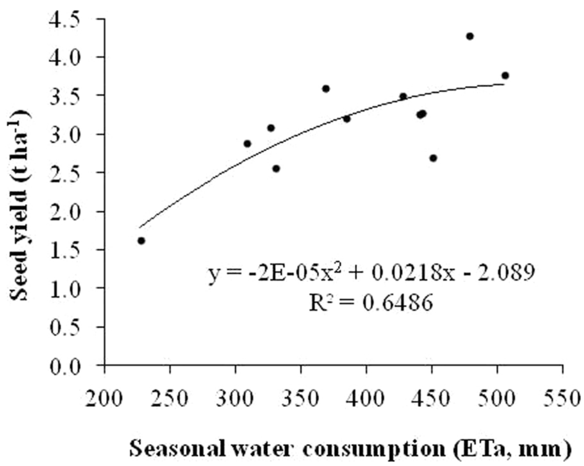

Fig. 2. Relationship between seed yield and seasonal water consumption (ETa).

(2005) and Kirnak et al. (2010) for semi-arid environment of west-central Nebraska, and the semi-arid Harran plain in Turkey, respectively. In Nebraska, Schneekloth et al. (1991) found a linear relationship between seed yield and ETa.

Most investigations pointed out a strong linear correlation between seasonal ETa and soybean yield. However, the slope of the regression line varied considerably between studies. Moreover, other forms of relationship (e.g. exponential, quadratic) between crop yield and ETa are reported. This may be attributed to the impact of different factors such as differences in seasonal precipitation amount, its frequency and temporal distribution, crop varieties, soil properties, adopted irrigation method and scheduling, and other weather parameters and agronomic management practices.

3.9. Effect of seasonal precipitation on soybean growth

The results of the present study indicated that seasonal precipitation and its distribution throughout the growing season play a significant role in irrigation needs, WP and yield of soybean. In the wet 2006 season, precipitation was nearly enough for achieving yields comparable to those with irrigation treatments. The longest dry period in 2006 was 19 days in August, but it was between two substantial rainfall events (25 and 20 mm). As such, an increase in crop ET by irrigation did not contribute to increase soybean seed yield, relative to the yield of the rain-fed control. Instead, the increase of crop ET was due to increased soil evaporation. In 2006 and 2007, the first water deficits were noted in the pre-flowering period (in all treatments). However, contrary to the wet year (2006), the amount of precipitation in the semi-dry year (2007) began to decrease in the pre-flowering period. In July and August of 2006, the amount of precipitation was 136 mm, whereas in 2007 it was only 71 mm. Therefore, in 2007, water deficit was compensated by irrigation.

In the dry year (2008), the amount of precipitation during the vegetative growth (May-June) of soybean was 76 mm – much lower than in the semi-dry year (2007) and, especially, in the wet year (2006), in which the amount of precipitation was 105 mm and 177 mm, respectively. For that reason, the irrigation in the driest year (2008) began 35 days earlier than in other two years; 120 mm of water was added in the I_{100} treatment and 50 mm in the I_{65} treatment. In 2008, the amount of precipitation for the remainder of the season was similar to that in 2007. In 2006, the irrigation depth of the treatments varied from approximately 39 to 60 mm. These irrigation depths accounted for 8–12% of the seasonal water requirement (ETa). In 2007 and 2008, irrigation accounted for 17–32% and 28–52% of the total seasonal ETa of soybean, respectively. In 2006, the final waterings in treatments I_{65} and I_{100} were applied around the middle of the third decade of July. In the semi-dry year (2007), the final waterings were applied at the beginning of the third decade of July and in the dry year (2008) in the first decade of August, in treatments I_{100} and I_{65}, and in the middle of the second decade of August in treatment I_{40}. The differences between irrigation treatments also included a difference in seasonal ETa values between the wet year (2006) and other two years. The seasonal ETa under deficit irrigation conditions varied among the treatments and growing seasons, depending on the total amount of water available to the crop and prevailing weather conditions (Table 3). The seasonal values of ETa of the treatments varied over a broad range: from 440 to 505 mm in 2006, 308–427 mm in 2007, and 227–450 mm in 2008. The presented data indicated that the difference between I_{100} and the non-irrigated treatment was the smallest in the wet year. It was only 65 mm. In 2007 and 2008, this difference was 119 and 223 mm, respectively. In addition, despite the fact that the growing season in 2006 was colder and wetter than in 2007 and 2008, the seasonal values of ETa of the different treatments were higher in 2006. The high ETa values in 2006 are attributable to increased evaporation, due to frequent wetting of the soil by precipitation. The results of the present study indicate that in such wet years no crop irrigation of soybean is needed in a temperate environment.

Fig. 3 shows the correlation between the above-ground biomass and ETa for all treatments in the dry years (2007 and 2008). The correlation (R^2 = 0.7641) established for the dry years is much stronger than when all the three study years are viewed together (R^2 = 0.4553).

Fig. 4 shows a good quadratic correlation (R^2 = 0.8573) between soybean seed yield and above-ground biomass in the dry growing seasons (2007 and 2008). However, the quadratic correlation is much weaker when the data on all the three growing seasons are combined (R^2 = 0.6459).

The soybean seed yield exhibited a quadratic correlation (Fig. 5) with the total seasonal amount of irrigation water, in the dry years (2007 and 2008), which implied that there is an optimal water application amount to maximize the effectiveness of irrigation water on increasing grain yield above rain-fed yields. The quadratic model in the present study also indicated that the soybean seed yield peaked at about 120 mm of overall seasonal irrigation, and that it began to decrease with larger seasonal amounts of water. Hence, in the dry years (2007 and 2008) ETa reached about 380 mm, which ensured the highest yield with irrigation of up to 120 mm. This means that for soybean grown in temperate conditions, two waterings of 60 mm each might be sufficient, and three of 40 mm are even better for high yields. This can be explained with good water holding capacity and moderate water permeability of medium textured, and well-structured deep soils of Vojvodina region. These amounts of irrigation water are also indicated by the observed data of WP and IWP, which were the highest in the I_{65}

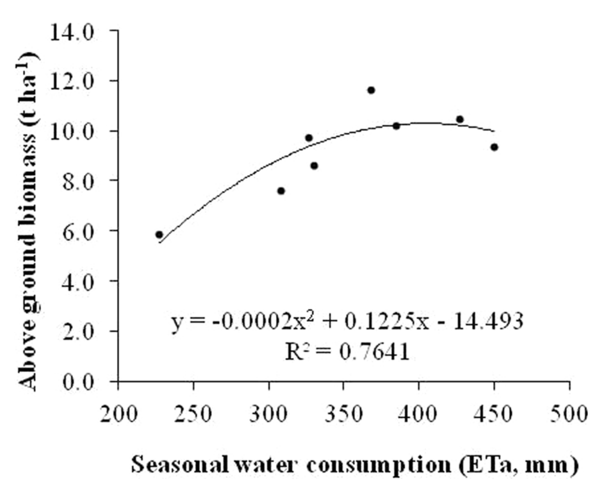

Fig. 3. Relationship between above ground biomass and seasonal water consumption (ETa).

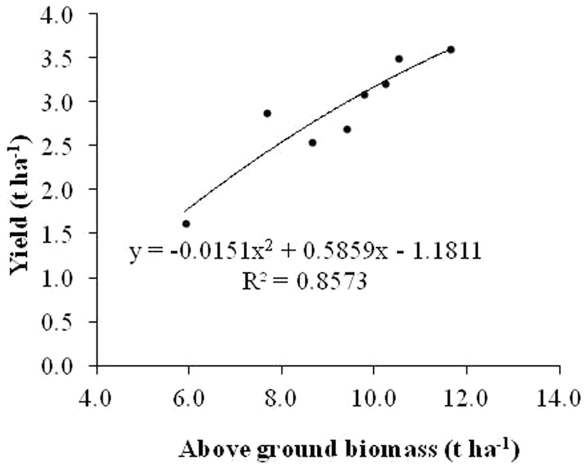

Fig. 4. Relationship between above ground biomass and seed yield for dry seasons (2007 and 2008).

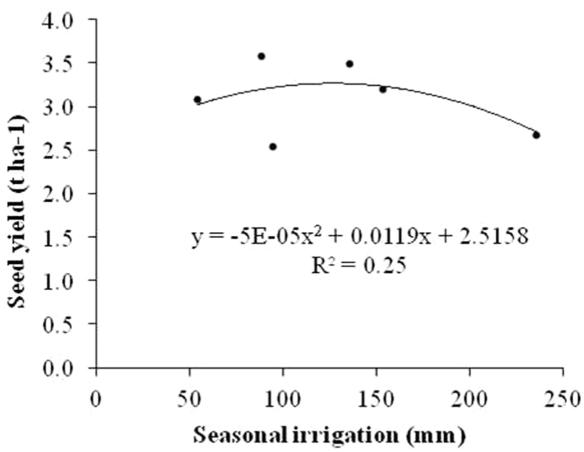

Fig. 5. Relationship between seasonal irrigation and seed yield for dry seasons (2007 and 2008).

treatment. Furthermore, that treatment exhibited the smallest WP and IWP deviations by year, which leads to production stability. If irrigation added more than 120 mm of water, evaporation increased but not the soybean seed yield.

4. Conclusion

The results of the three-year study indicated that irrigation regimes significantly affected soybean seed yield and WP in different grades and depending on seasonal precipitation. A quadratic relationship between seed yield and seasonal crop ETa was observed for the three-year data. As a rule of thumb, a sum of precipitation and irrigation should reach about 380 mm during the growing season to guarantee high yield and WP of soybean. Furthermore, the results showed that the amount of precipitation in two dry growing seasons (2007 and 2008) was insufficient and that irrigation was needed to increase and stabilise yield. However, in the wet year (2006), an increase in the crop ETa due to irrigation did not contribute to a significant increase of soybean yield in respect to the non-irrigated treatments. This is because the share of evaporation in the total ET increased. The results of the study suggested that in such wet years no soybean irrigation is needed in a temperate environment. With regard to dry years, when irrigated ETa reaches about 380 mm, the yields are the highest when up to 120 mm of irrigation water is added. Irrigation of more than 120 mm increases soil evaporation but does not increase the yield.

Deficit irrigation treatments increased aboveground biomass, seed yield and other investigated variables compared to non-irrigated treatment. Soybean crops under rain-fed conditions give highly variable yield, making irrigation strongly required in temperate climates. The highest yield was obtained from the I_{65} treatment in all three growing seasons, which indicates that deficit irrigation is good strategy in soybean production in terms of yield maximization and water productivity. To optimize irrigation management, an economic analysis is required. It depends on the objectives of irrigation, whether the objectives are related to maximization of profits, water productivity or yield, or to optimize among three objectives, which might be a case study. Nevertheless, satisfactory seed yield was obtained with irrigation level I_{40}. Although the I_{40} plot received about 255% less irrigation water compared to the I_{100} treatment, seed yield was reduced by an average of 20% compared to I_{100}. Therefore, the I_{40} irrigation strategy proved to be efficient in saving water under conditions of restricted water resources. Results indicated that crop water productivity was the highest with irrigation regime I_{65} as shown by the WP and IWP values, indicating that this irrigation regime contributes to the most stable soybean production. The economic analysis between I_{65} and I_{40} deficit irrigation treatments is necessary to be done in order to decide which one of those two treatments is more efficient. In addition to this, irrigation timing in soybean production is very important to be performed in sensitive phenological stages, which directly promote regulated deficit irrigation as a tool to enhance overall irrigation management.

Acknowledgement

This work was supported in part by the Ministry of Education, Science and Technological Development of the Republic of Serbia (Contract No. III 43009).

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